

AI BOOTCAMP • MAVEN

CloudSync

Multi-Agent Support System

Week 1 Assignment — OpenAI Agent Builder

Christian Bobadilla

ChatKit

Agent Builder

FastAPI

THE CHALLENGE

Why a Single Model Fails

CloudSync handles 15,000 support tickets/month with varying complexity. Using one LLM for everything either overspends on simple tickets (GPT-5.2 at \$0.05/call) or fails on complex ones (GPT-5-nano at \$0.001/call). The assignment: build a routed multi-agent system that matches model cost to ticket complexity.



60%

Simple

FAQ lookups

GPT-5-nano · \$0.001/call



25%

Medium

Billing & account issues

GPT-5-mini · \$0.01/call



15%

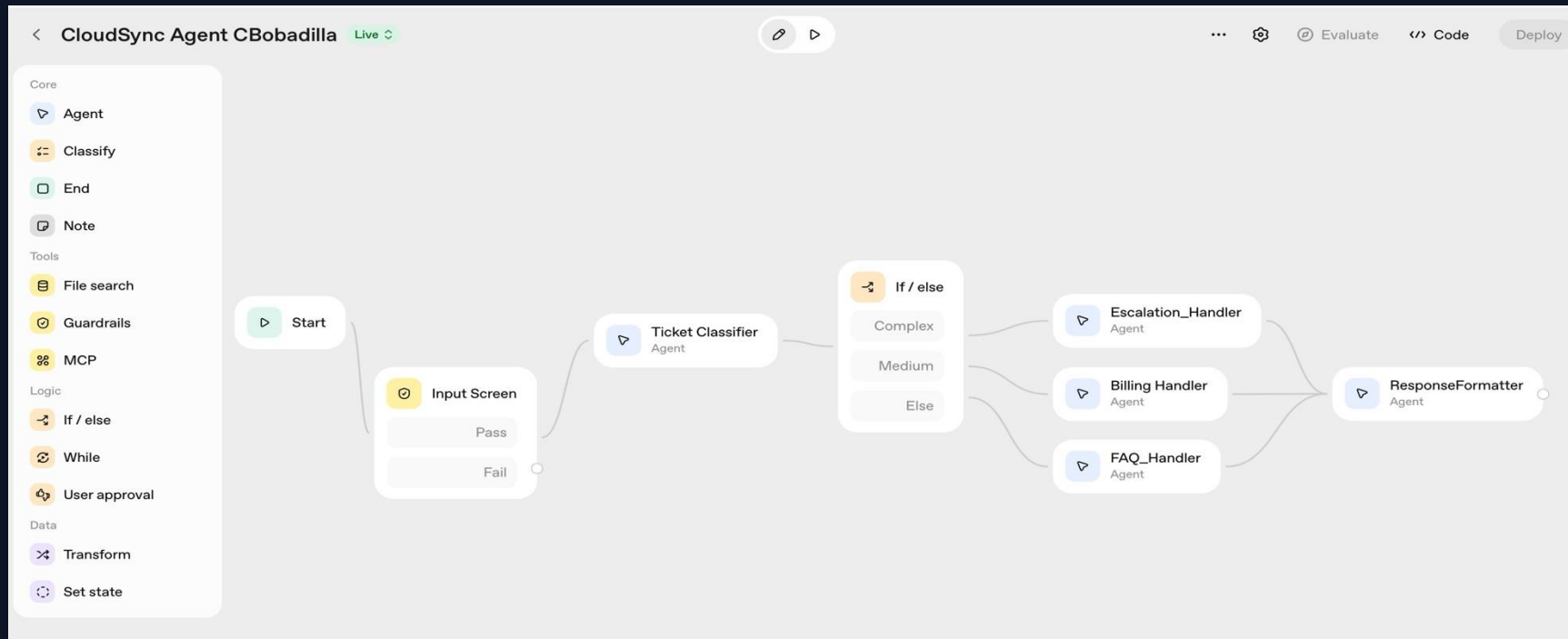
Complex

Technical failures, churn risk

GPT-5.2 · \$0.05/call

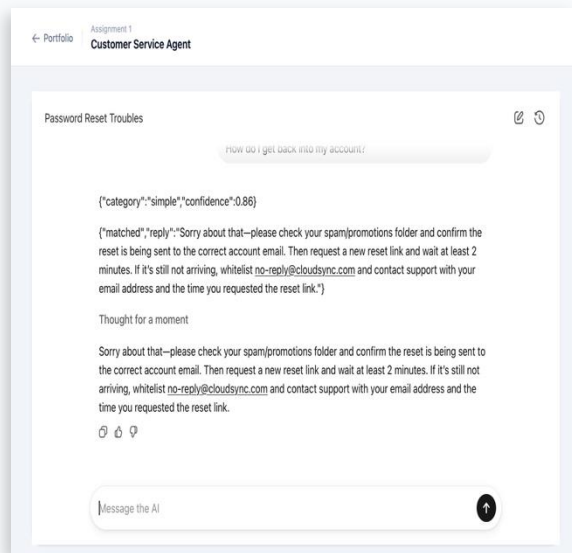
DELIVERABLE 1

Working Multi-Agent System



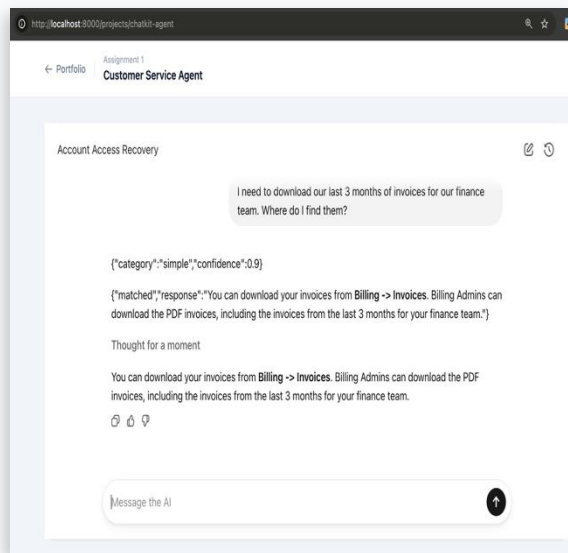
DELIVERABLE 2

Sample Tickets – Simple & Medium



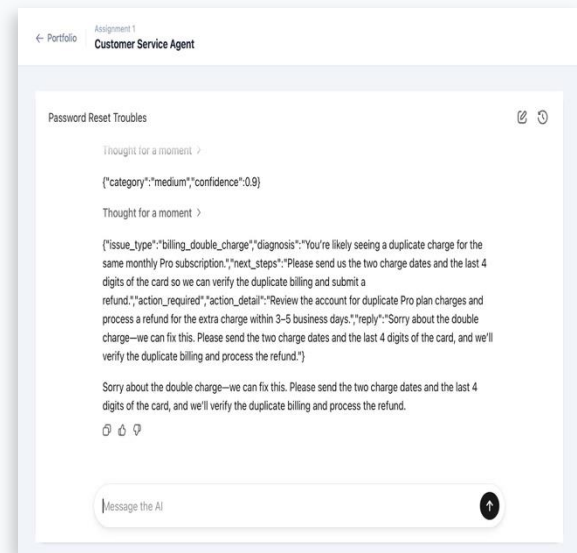
simple • 0.86

Ticket 1: Password Reset



simple • 0.9

Ticket 2: Invoice Lookup



medium • 0.9

Ticket 3: Double Charge

DELIVERABLE 2

Sample Ticket – Complex Escalation

[← Portfolio](#) | Assignment 1
Customer Service Agent

Password Reset Troubles

Thought for a moment

I'm sorry—this has become a repeated pattern, and I understand why leadership is reviewing alternatives. We'll take ownership and coordinate resolution immediately.

For the current syncing issue, we're opening an urgent technical escalation now. For last month's incorrect billing, we'll audit the account and correct the charge if confirmed. For the access loss incident, we'll review authentication/seat/access logs to identify the root cause.

To ensure someone is accountable end-to-end, please reply with your account email and (if available) the dates/times of:

1. the incorrect billing,
2. the access outage,
3. the current sync failure.

Then we'll assign this to an incident owner internally and provide you with an updated status timeline.

Message the AI

complex · escalation

Ticket 5: Considering Cancellation

Third issue this quarter: incorrect billing, team lost access, data not syncing. Leadership reviewing alternatives.

Escalation Handler (GPT-5.2)

- Acknowledged the pattern and validated frustration
- Committed to ownership and immediate coordination
- Opened urgent technical escalation for sync issue
- Initiated audit for billing and access incidents
- Requested specific dates/times for all 3 incidents
- Promised an incident owner and status timeline

Architecture Explanation

Model Choices

Ticket Classifier: GPT-5-nano (\$0.001) — lightweight pattern recognition, outputs JSON {category, confidence}

FAQ Handler:
GPT-5-nano (\$0.001) — simple lookup against FAQ knowledge base

Billing Handler:
GPT-5-mini (\$0.01) — medium complexity, account & billing logic

Escalation Handler:
GPT-5.2 (\$0.05) — complex reasoning for high-risk tickets, SLA-aware

Response Formatter:
GPT-5-nano (\$0.001) — standardizes output tone

Routing Logic

Input Screen: Guardrails validate ticket structure before processing

Classifier Agent:
Reads ticket, outputs structured JSON with category + confidence score

If/Else Router:
Complex → Escalation Handler, Medium → Billing Handler, Else → FAQ Handler

Low confidence:
Uncertain tickets (< 0.7) route to Escalation for safety

All paths converge:
ResponseFormatter ensures consistent output

Why This Wins

Cost savings: ~68% cheaper than GPT-5.2 for all tickets (\$0.0134 vs \$0.05 blended avg)

Quality match:
Complex tickets still get GPT-5.2's full reasoning power

Reliability:
Simple tickets handled by proven FAQ lookup, not overkill LLM

Scalability:
Add new categories/handlers without changing the classifier

Safety:
Guardrails + confidence routing prevent misclassification from reaching users

Get the Code & Try It Live



GitHub Repository

github.com/cabobadilla/AgenticVault-ChrisBob/tree/main/Assignment1



Live Project — Railway

agenticvault-chrisbob-production.up.railway.app